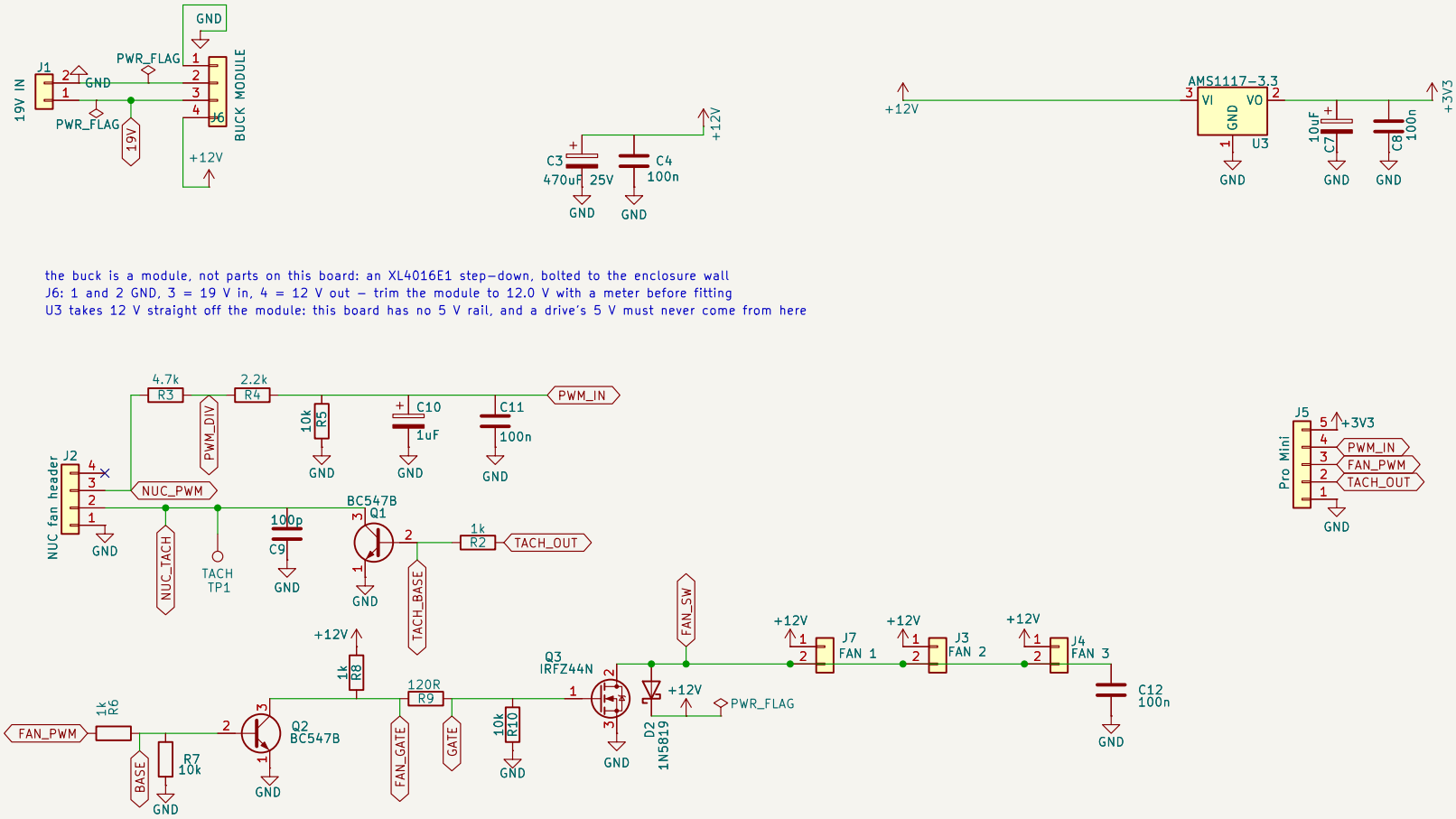

19V IN
J1
GND
PWR_FLAG
19V
BUCK MODULE
J6
+12V
C3
470uF 25V
C4
100n
AMS1117-3.3
VI
VO
U3
10uF
C7
C8
+3V3
the buck is a module, not parts on this board: an XL4016E1 step-down, bolted to the enclosure wall
J6: 1 and 2 GND, 3 = 19 V in, 4 = 12 V out — trim the module to 12.0 V with a meter before fitting
U3 takes 12 V straight off the module: this board has no 5 V rail, and a drive's 5 V must never come from here
4.7k
R3
2.2k
R4
PWM_DIV
10k
R5
C10
1uF
C11
100n
PWM_IN
J2
NUC fan header
NUC_PWM
NUC_TACH
TACH
TP1
100p
C9
BC547B
Q1
1k
R2
TACH_OUT
TACH_BASE
J5
Pro Mini
FAN_PWM
TACH_OUT
FAN_SW
Q3
IRFZ44N
D2
1N5819
J7
FAN 1
J3
FAN 2
J4
FAN 3
C12
100n
R8
120R
R9
R10
FAN_GATE
GATE
R6
R7
BASE
Q2
BC547B
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